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Synthesis and application of isoindigo quaternary ammonium salts in copper plating

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Achieving uniform through-hole filling by copper electrodeposition is important for the fabrication of printed circuit boards (PCBs), and the development of suitable additives is a key process to realize efficient electrodeposition. Five isoindigo (IID) quaternary ammonium salts were synthesized using isoindigo molecules as the core skeleton. Electrochemical tests, theoretical calculations, and molecular dynamics simulations demonstrated that the isoindigo derivatives have the ability to act as leveler and exhibit electrode absorption capacity, among which IID-C4-QL showed the best inhibition of copper deposition as well as the best cathodic adsorption ability. The high through-hole filling and excellent leveling performance of IID-C4-QL were confirmed by actual electroplating tests.

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1. Introduction

With the continuous development of the information age, electronic devices are being more widely used, and high-density interconnection (HDI) printed circuit boards (PCBs) are one of the key components in the preparation of electronic products,[1,2] in which through-hole (TH) metallization is a pivotal technology for PCB electrical interconnection. Considering the economic cost and the performance advantages of copper metal such as good electrical conductivity and high electromigration resistance, copper electrodeposition is mostly used to realize uniform through-hole filling.[3–5] With the increasing technological demands in the field of electronics manufacturing, PCBs also need to develop in the direction of high density and high integration,[6] and achieving a higher level of through-hole filling presents a great challenge for the development of copper electrodeposition.[7,8]

In order to achieve the bottom-up filling effect without defects in the through-hole, the main option is to add functional additives during copper deposition,[9] which are usually classified into three categories according to the functions they perform in copper plating: inhibitors, levelers and accelerators.[10–13] Inhibitors are mainly polyether compounds, such as polyethylene glycol (PEG), polypropylene glycol (PPG), and most of the large organic molecules with their molecular weights in the range of 6000–10 000, which mainly inhibit the deposition of copper.[14,15] The most common accelerator is sodium polydithiodipropane sulphonate (SPS), which is able to synergize with chloride ions to increase the rate of copper deposition and refine the copper crystalline particles, resulting in a bright copper coating.[16,17] Levelers are the most complex of the additives and a special kind of inhibitor, which can further enhance the dispersing ability of the plating solution and the flatness of the copper layer.[18,19] A series of dye-based levelers with large conjugation planes, such as diketopyrrolopyrrole derivatives, anthraquinone derivatives, 3,3′-bicarbazole derivatives, *etc.*, have been reported in previous works of our group,[2,20,21] and it is hypothesized that molecules possessing good planarity as well as large conjugation are better at leveling.

Based on the preliminary work of the group, isoindigo organic functional materials were selected to explore their feasibility as levelers. The isoindigo molecule is one of the three isomers of the indigo pigment, which has a highly conjugated large Π-bond structure, strong planarity and symmetry. In this paper, a series of isoindigo quaternary compounds (IID-C4-QL, IID-C4-Bpy, IID-C4-py, IID-C4-IMD, and IID-C4-DMAP) were synthesized *via* a two-step reaction of alkylation and quaternization with the isoindigo molecule as the core skeleton and the side chain modification of the nitrogen atoms on the lactam ring.

The synthesized isoindigo quaternary ammonium salts (IID-QAS) were electrochemically tested to evaluate their inhibitory performance and convective adsorption, and the energy band gap and adsorption energy of five quaternary ammonium salts were calculated by combining quantum chemical calculations and molecular dynamics simulations to analyze their adsorption behaviors on the surface of copper, and the better quaternary ammonium salts were screened out from them for the next step of the actual copper plating test. The TP values were measured by making metallographic gel of copper plated through holes, and the morphology of the copper-plated

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surface was analyzed by SEM and XRD, which was corroborated *via* the electrochemical test. The results showed that IID-C4-QL had outstanding inhibition performance in the electrochemical test, and had a leveling effect in the actual copper-plating, which visually improved the TP value, refined copper grain size on the copper surface, and made the surface bright and flat, and the optimum amount of the quaternary ammonium salt was 1.5 ppm. The use of isoindigo derivatives as novel levelers broadens the application scope of isoindigo dyes and lays the foundation for exploring other conjugate structures that can be used as electroplating levelers.

2. Experiment

2.1. Synthesis and characterization of materials

Since isoindigo is not yet commercially available in industrial and chemical grades as a feedstock, five isoindigo quaternary ammonium salts IID-C4-QL, IID-C4-Bpy, IIDC4-py, IID-C4-IMD and IID-C4-DMAP were synthesized using 2-indolone and indigo as raw materials for the synthesis of isoindigo matrices, and five IID-QAS were synthesized using two-step alkylation and quaternization, and the five II-QAS structural formulas are shown in Fig. 1 and the synthetic route is described in Scheme 1.

2.1.1. Synthesis of isoindigo materials. The reaction solution was mixed with indigo red (1 mmol), 2-indolone (1 mmol), 1 mL of dilute hydrochloric acid as catalyst and 20 mL of acetic acid, which reacted in a 100 mL three-necked round-bottomed flask at 120 °C for 4 h, and was then cooled to room temperature. The reaction solution was removed and filtered, and 10 mL of cold methanol solution, 10 mL of deionized water, 10 mL of aqueous sodium bicarbonate solution and 10 mL of cyclohexane were used to remove impurities such as acetic acid and raw materials that had not reacted completely, and then the dark-red raw material product was obtained after drying, and the yield was more than 95%, so it could be used in the next step of the reaction directly.

2.1.2. Synthesis of isoindigo alkylation products. IID (1 mmol), potassium *tert*-butoxide (4 mmol), 20 mL of the DMF solvent were added into a 100 mL three-necked round-bottomed flask, under the protection of argon at 40 °C, first reacted half an hour and then the alkylation reagent 1,4-dibromobutane (10 mmol) through the pinhole syringe. The reaction temperature was increased to 60 °C, the reaction was continued for 6 hours and then cooled to room temperature. The reaction solution was added into a dispensing funnel with 15 mL of water, and then extracted with dichloromethane several times to remove the solvent DMF. The upper layer of the liquid was counted into a 100–200 mesh silica powder, evaporated to make a dry sample, and then the product was purified *via* column chromatography on silica columns (200–300 mesh) using ethyl acetate and petroleum ether (10 : 1, v/v) as the eluents; eventually the dark-red solid product IID-alkylated product was obtained with a yield of 43–60%.

2.1.3. Synthesis of isoindigo quaternized products. The alkylated product of isoindigo (0.5 mmol), and five quaternary ammonium reagents, namely pyridine, bipyridine, quinoline, vinylimidazole, and 4-dimethylaminopyridine (5 mmol), were added into a 120 mL acetonitrile tube with 20 mL acetonitrile solvent, and then heated to 80 °C for 12 h. The reaction solution was cooled to room temperature, and then poured into ethyl acetate, and led to the precipitation of the reddish-brown solid. The quaternary ammonium salt product IID-QAS was obtained by filtration and drying with a yield of 87–95%.

2.2. Electrochemical analysis

In this paper, five synthesized isoindigo quaternary ammonium salts were electrochemically characterized *via* cyclic voltammetry (CV) measurements, potentiodynamic polarization, and galvanostatic measurements (GMs). A three-electrode system (rotating disc electrode for the working electrode, Ag/AgCl

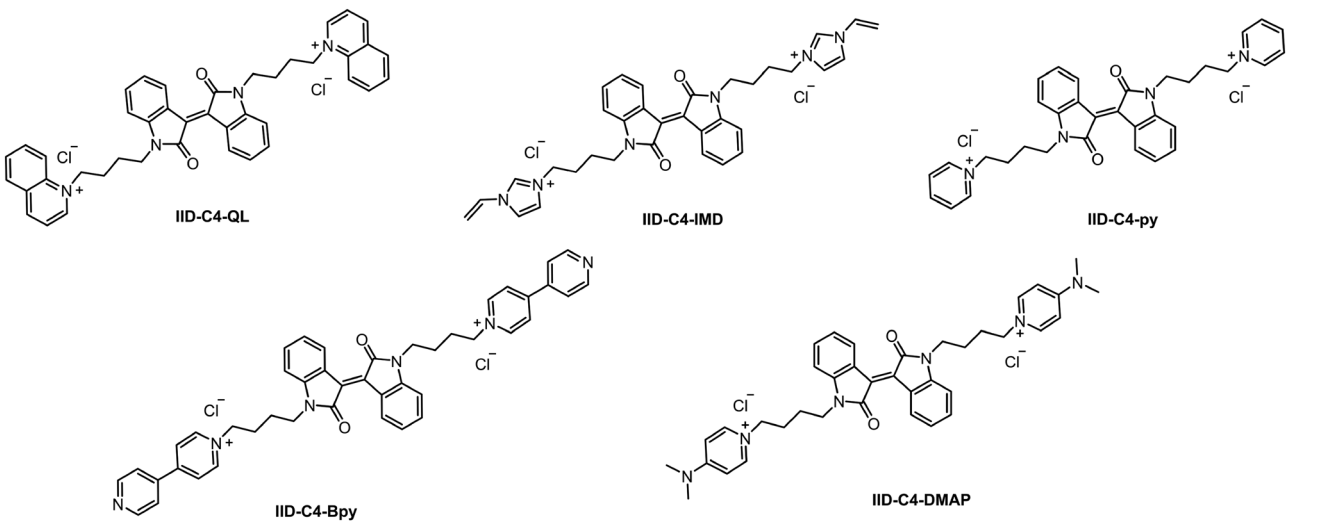
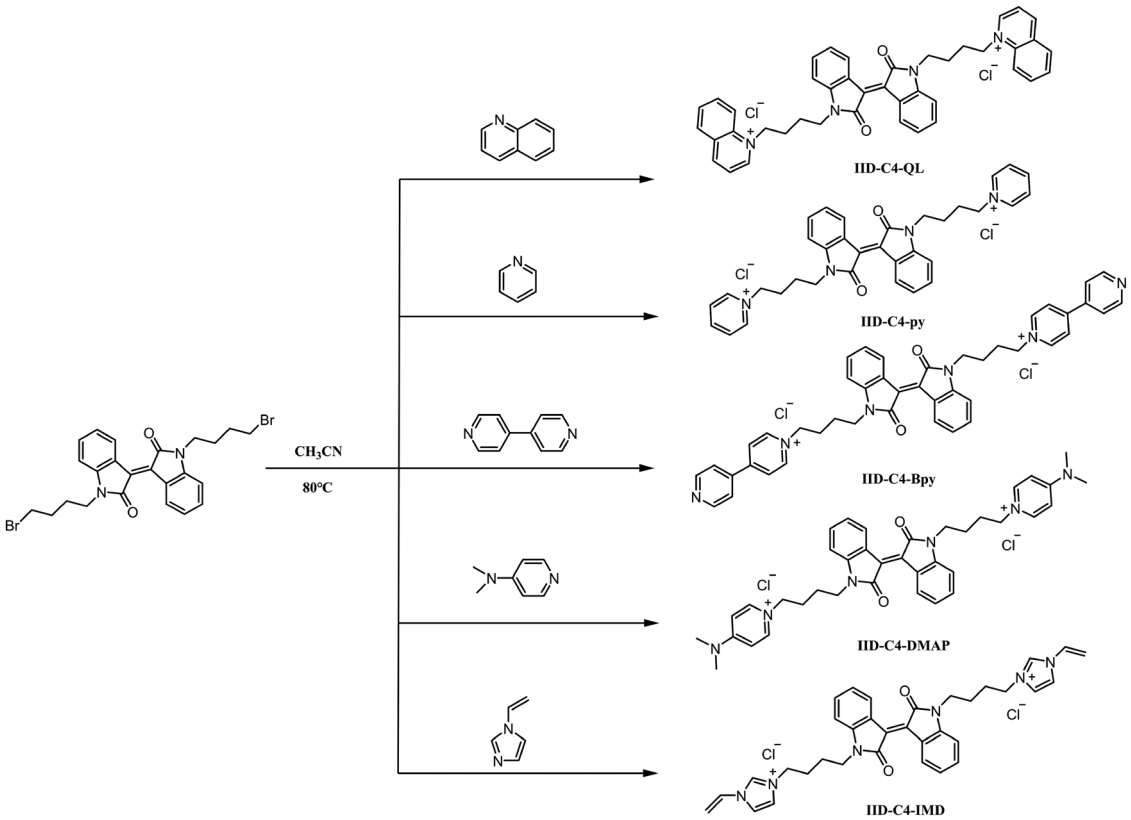


Fig. 1 Molecular structure of five IID quaternary ammonium salts.

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Scheme 1 Synthesis route of IID-QAS levelers.

electrode for the reference electrode, and Pt electrode for the auxiliary electrode) was used for the electrochemical tests, and the virgin make-up solution (VMS) used in the tests was composed of 200 g L^{-1} H_2SO_4, 60 g L^{-1} $CuSO_4{\cdot}5H_2O$, and 50 ppm Cl^-. In the cyclic voltammetry test, the scanning potential was in the range of 1.57 to −0.25 V, and the constant scanning rate was 50 mV s^{-1}. The polarization curve test was performed with 120 s of electrode pretreatment before the test, and the copper electrodeposition was carried out for a short period of time in a blank plating solution, with the scanning point set from 0 to −0.8 V and the scanning rate at 20 mV s^{-1}. Galvanostatic measurements set the high speed to 1500 rpm s^{-1} and the low speed to 150 rpm s^{-1}, and testing was performed for 4000 s at a constant current of 3 mA.

2.3. Calculations and simulations

Theoretical calculations are based on density-functional theory (DFT) using Gaussian software, the specific parameters of the B3LYP calculation method and the basis group of 6-31G. The molecular structure of the quaternary ammonium salt needs to be optimized before calculations are performed to obtain molecular energy level orbital data and relevant electrostatic potential distributions. The specific HOMO/LUMO orbitals and the energy level gaps are derived using the software Multiwfn.[22,23] The molecular dynamics simulation was performed using the software Material Studio, and the simulation modeled a copper cell with a crystal surface index of (111), a thickness of 7.0, a cell boundary of 16 nm in length and width, a quaternary ammonium molecule of 1 and a water molecule of 2000, a simulated temperature of 298 K, a force field of COMPASS II, with fine precision.

2.4. Electroplating measurements

Through-hole plating experiments were carried out in a Haring cell with 1.5 L VMS; copper phosphorus-containing plates were inserted on both sides of the bath as the anode, and PCBs were inserted in the middle of the bath and secured with clips as the cathode to perform copper electrodeposition; a PCB thickness of 2 mm with multiple *via* holes with a radius of 150 μm. 200 ppm PEG, 1 ppm SPS, and various concentrations of IID-QAS were added into the plating solution. Before plating, the PCB should be treated in advance, using a degreaser and acid cleaner to remove the oxidized protective film on the surface, and deionized water to remove the residual agent on the surface. The PCB was pre-plated for 10 min to remove impurities in the plating solution, and then the treated PCB was used for plating, and the current was controlled to maintain a current of 1.5 A for 70 min of copper plating. Scanning electron microscopy (SEM) was used to observe the surface morphology of PCB slices, and X-ray diffractometry (XRD) was used to analyze the crystal orientation. Metallographic microscopy was used to observe the thickness of the copper plating layer of the through holes and calculate the TP value, and TP value was calculated as shown in Fig. 2.

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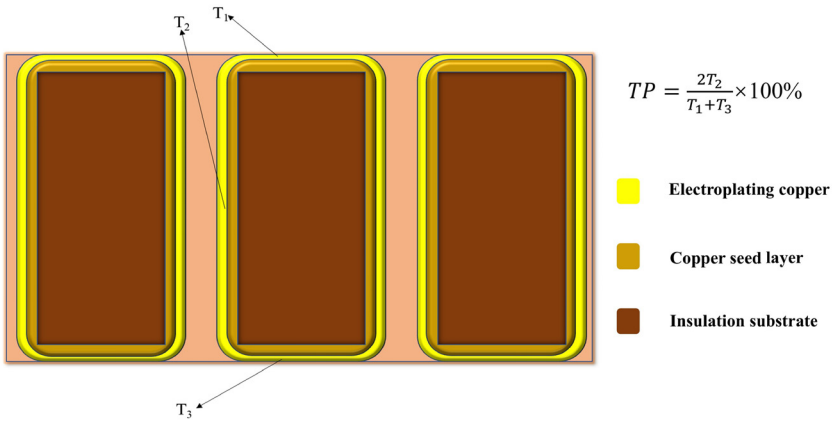


Fig. 2 PCB through-hole diagram and TP value calculation formula.

3. Results and discussion

3.1. Cyclic voltammetry measurements

Cyclic voltammetry is one of the commonly used methods for analyzing the inhibitory effect of additives on metal electrodeposition.[24] The negative peak in the curve represents the cathodic reduction peak of copper ions, while the positive peak represents the anodic stripping peak of copper. The peak areas in the copper deposition zone and the copper stripping zone correspond to the deposition amount of copper ions and the stripping amount of metallic copper, respectively, and these two quantities are directly proportional.[25] Therefore, the deposition of copper ions is represented by the positive peak area, and the smaller the positive peak area, the stronger the inhibitory effect of the additive on copper ion deposition.[26] The results in Fig. 3 show that at a concentration of 2 μmol L^{-1}, the peak areas of all six additives (including JGB) decreased compared to VMS, indicating that all additives have inhibitory effects. Table 1 calculates the peak areas of all positive peaks

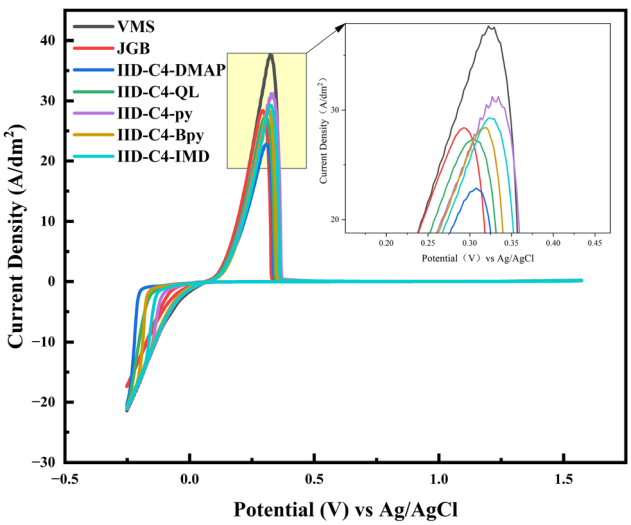


Fig. 3 CV curve of JGB and IID-QAS at the same concentration (2 μmol L^{-1}).

Table 1 Peak area of six additives and VMS

	VMS	JGB	DMAP	QL	Py	Bpy	IMD
Peak area	4.72	3.19	2.46	3.09	3.85	3.14	3.51

using integration, where the peak areas of IID-C4-QL, IID-C4-DMAP, and IID-C4-Bpy are all smaller than that of JGB.

The three IID-QAS with the best inhibitory performance (better than JGB) were selected to test the effect of concentration change on the inhibition ability and compared with the cyclic voltammetry curves of JGB as shown in Fig. 4. The maximum current densities are decreasing with the increase of the concentration, among which IID-C4-DMAP and IID-C4-QL show the largest decrease which the maximum current density decrease to below 5 A dm^{-2} at above 8 μmol L^{-1}. It can be concluded that the two quaternary ammonium salts, IID-C4-DMAP and IID-C4-QL, have the best inhibitory performance and the improvement of inhibitory performance under the change of concentration is also the most obvious, which indicates that these two IID-QAS have the inhibitory ability to be levelers.

3.2. Polarization measurements

The polarization curve can analyze the corresponding change of copper electrodeposition on the cathode surface, and the potential of the transverse coordinate of the folding point in the curve where the current density rises rapidly is the reduction potential of copper ions.[27,28] A larger absolute value of the reduction potential represents a decrease in the rate of deposition, indicating that the compounds are more inhibitory, which can be combined with cyclic voltammetry curves to mutually confirm the inhibitory ability of the different IID-QAS. Fig. 5 shows that the reduction potential of the VMS is about −0.075 V, and the addition of JGB only reduces to about −0.08 V, which is a relatively small change. The reduction potentials of IID-C4-IMD and IID-C4-py are significantly higher than that of JGB, but they are also only before −0.1 V. However, IID-C4-DMAP and IID-C4-QL, which exhibit well-defined cyclic voltammetry

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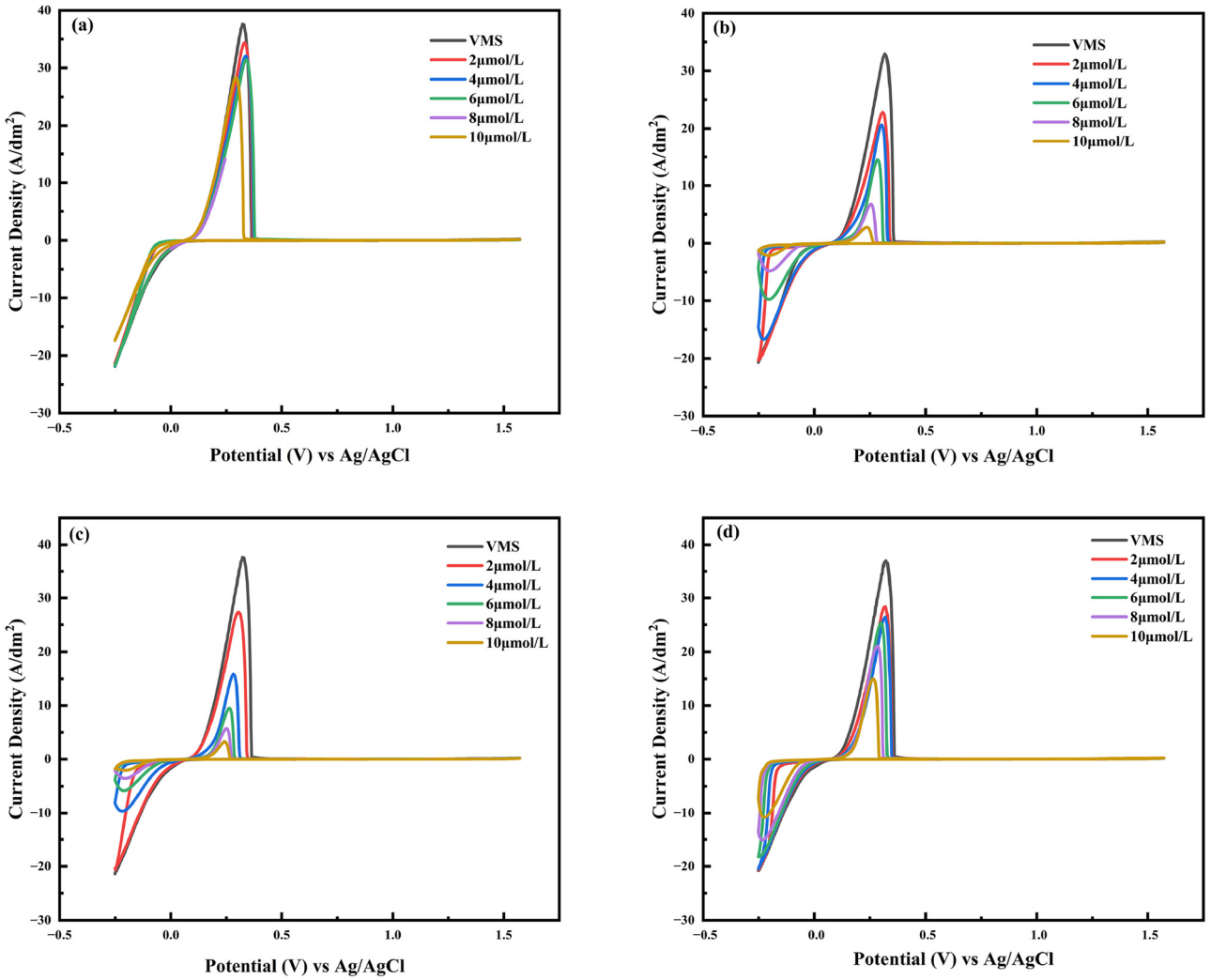


Fig. 4 (a) CV curve of JGB at different concentrations; (b) CV curve of IID-C4-DMAP at different concentrations; (c) CV curve of IID-C4-QL at different concentrations; (d) CV curve of IID-C4-Bpy at different concentrations.

curves, also show significant activity in polarization curves. Their reduction potentials shifted to −0.166 V and −0.171 V, respectively, consistent with the cyclic voltammetry data. The addition of IID-C4-DMAP and IIDC4-QL can help copper ions to be deposited more slowly, indicating that the inhibitory effect of these two IID-QAS is strong.

As shown in Fig. 6, the polarization curves of IID-C4-DMAP and IID-C4-QL at different concentrations indicate that the reduction potential decreases with increasing concentration. The magnitude of this change aligns closely with the concentration gradient. At the addition of 10 μmol L^{-1} of IID-QAS, the reduction potential is after −0.20 V and there is a tendency to decrease with the increase of concentration. Above all, IID-C4-DMAP and IID-C4-QL exhibit much stronger inhibitory performance than JGB, which is in line with the electrochemical properties required for the leveler.

3.3. Galvanostatic measurements

Galvanostatic measurements were performed for the IID-C4-DMAP and IID-C4-QL, using a rotating disk electrode speed of 1500 rpm to simulate the strong convection zone at the copper through-hole orifice and 150 rpm to simulate the weak convection zone inside the hole.[29] PEG, SPS and IID-QAS were added sequentially to the VMS solution, and the potential changes were observed in real time to determine the effect of the additives on the system. As shown in Fig. 7, $\Delta\eta = \eta 150\ \text{rpm} - \eta 1500\ \text{rpm}$ represents the potential difference between high and low convection speeds, and a larger $\Delta\eta$ indicates that the additives are more affected by the convective effect, which is more favorable for the internal deposition of the through-hole.[30,31] There was a sharp decrease in potential when 200 ppm PEG was added at 1000 s, and a significant increase in potential when 2 ppm SPS was added at 2000 s, which are consistent with the effects of inhibitory and accelerated abilities of inhibitors and accelerators on the potential of Cu ions deposition, but the potential difference of the curves is less than 1–2 mV, indicating that they are not affected by the effects of convection. The addition of ID-C4-DMAP and IID-C4-QL at 3000 s produced a potential difference, which the potential at low rotational speeds

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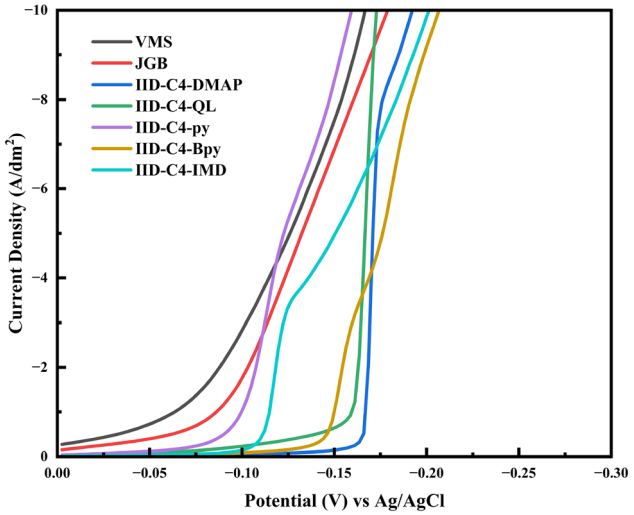


Fig. 5 Polarization curve of JGB and IID-QAS at the same concentration (2 μmol L^{-1}).

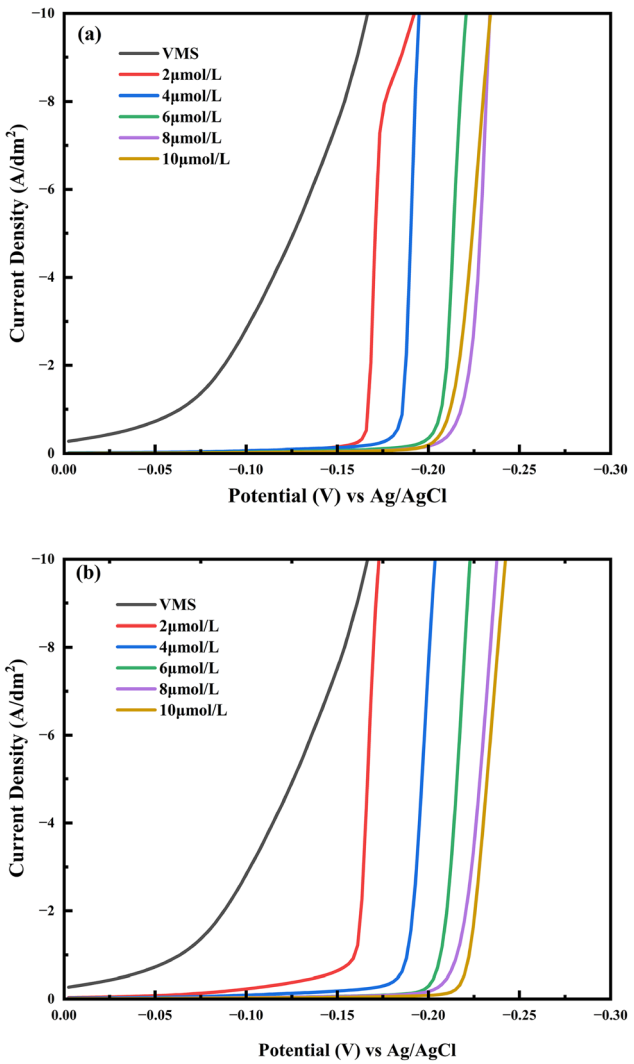


Fig. 6 (a) Polarization curves of IID-C4-DMAP at different concentrations; (b) polarization curve of IIDC4-QL at different concentrations.

was higher than that at high rotational speeds, suggesting that at high rotational speeds the inhibitory effect of the two additives was stronger, and more adsorbed at the pore openings, with convection-dependent adsorption (CDA), which is an essential property for levelers.

Comparison of the two additives reveals that the potential difference for IID-C4-DMAP is $\Delta\eta_{DMAP} = \Delta\eta_4 - \Delta\eta_3 = 5$ mV, whereas the potential difference for IID-C4-QL is $\Delta\eta_{QL} = \Delta\eta_4 - \Delta\eta_3 = 14$ mV, which is larger, and it can be concluded that the capacity of absorption of ID-C4-QL at the pore opening is stronger.

In the GMs, IID-QAS still shows obvious inhibition after the addition of PEG and SPS, indicating that IID-QAS can work well in the plating system with PEG and SPS and chloride ions, and the addition of IID-QAS do not bring the potential back to that of the initial VMS, so it is assumed that there is a synergistic effect between IID-QAS and other additives in the system. Moreover, when SPS and IID-QAS exist at the same time, there is a certain competitive effect of absorption between them. Fig. 7 shows an increasing trend of potential at 3000 s, which indicates that the accelerating effect of SPS still exists, but after the addition of IID-QAS, the potential drops steeply, and the accelerating effect of SPS is weakened due to the competitive adsorption of IID-QAS; the more the IID-QAS added, the more the IID-QAS is able to dominate in the competitive adsorption, and thus the overall plating system shows inhibitory effect.

3.4. Quantum chemical calculations

Quantum chemical calculation is used to analyze the correlation between the molecular structure and the electronic properties of the five IID-QAS and to analyze the effect of different molecular structures on copper plating.[32] According to the frontier molecular orbital theory, the energy of the highest occupied molecular orbital (HOMO) is usually related to the electron-giving ability of the molecule and the energy of the lowest unoccupied molecular orbital (LUMO) is related to the electron-absorbing ability of the molecule. The difference of the energy band gap (ΔE) is defined as follows:

$$\Delta E = E_{HOMO} - E_{LUMO} \quad (1)$$

The smaller the energy level difference, the more stable the molecule is for adsorbing the copper layer.[33] From Fig. 8 and Table 2, the order of ΔE of the five IID-QAS energy level differences is $\Delta E_{QL} < \Delta E_{DMAP} < \Delta E_{Bpy} < \Delta E_{py} < \Delta E_{IMD}$, the one with the smallest energy level difference is IID-C4-QL, which shows that its stability is the best in adsorbing on the copper layer, and at the same time, it is the highest molecule in terms of E_{HOMO}, which means that the molecule has the strongest electron-donating ability. The oxygen and nitrogen atoms on the lactam ring are the main components of the HOMO orbital, so they are easily come under the electrophilic attack in the conjugated system and the IID-C4-QL of the high E_{HOMO} is easily superimposed with the d orbitals of the metal, so it is assumed that the lactam ring of the isoindigo can help the molecule to combine with the copper, occupying the

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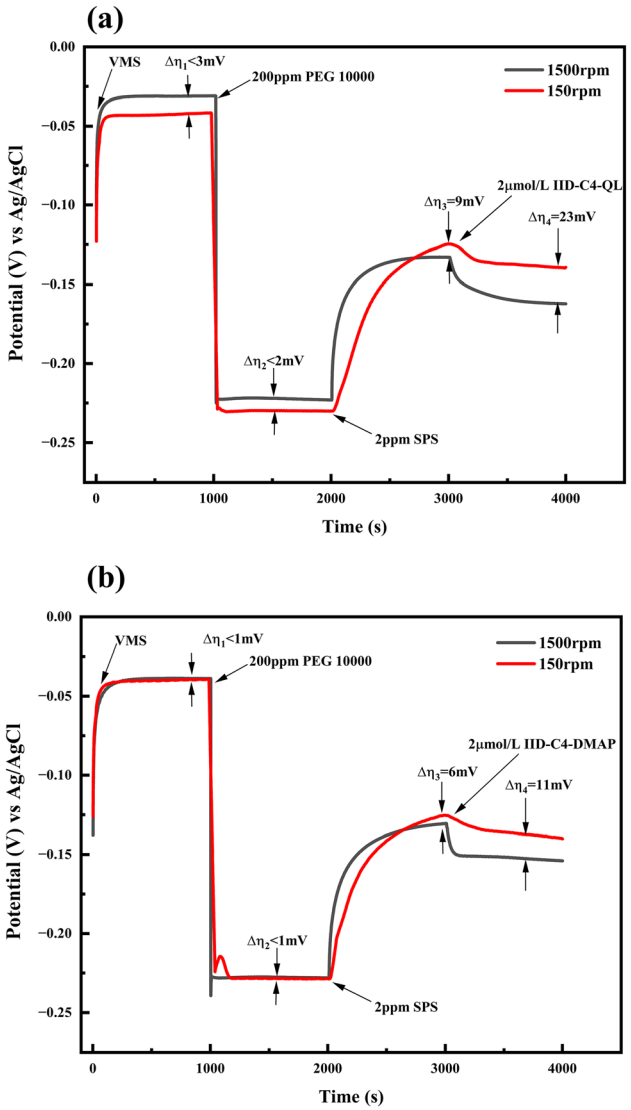


Fig. 7 (a) Galvanostatic measurement curves of IID-C4-DMAP at 150 rpm and 1500 rpm; (b) galvanostatic measurement curves of IID-C4-QL at 150 rpm and 1500 rpm.

adsorption site of copper ions, and thus effectively inhibiting the precipitation of copper in the electrodeposition.

Fig. 9 shows the electrostatic potential distribution of IID-QAS, where red represents high electrostatic potential regions, primarily distributed at the terminal quaternary ammonium groups, and blue represents low electrostatic potential regions, primarily distributed within the lactam ring of the molecule.[34,35] Based on the conclusions from the molecular HOMO/LUMO orbitals, the internal lactam ring in the molecule serves as the reaction site for copper electrodeposition, while the quaternary ammonium groups at the terminal ends act as positively charged groups that physically adsorb onto the electrode surface under the influence of an external electric field. With the combined effects of these two mechanisms, IID-QAS adsorbs onto the copper surface, thereby inhibiting copper ion deposition. Among them, IID-C4-QL has the highest electrostatic potential energy and the strongest physical adsorption capacity.

Molecular dynamics simulation (MD) is used to study the adsorption behavior of five quaternary ammonium salts on the copper surface. Fig. 10 shows the process and final morphology of the molecules adsorbed on the copper surface, and the adsorption energy (E_{ads}) of the molecules was obtained by calculating the difference between the total and partial energies of the adsorption at the completion of steady state,[36] with the following equation:

$$E_{ads} = \frac{1}{N(E_{total} - E_{Cu} - N \times E_{molecule})} \quad (2)$$

where N is the number of molecules adsorbed in the simulation, E_{total} is the total energy of the final adsorption, E_{Cu} is the energy of the initial layer of copper atoms, and $E_{molecule}$ is the energy of the leveler molecules alone. In the simulation of adsorption, the number of molecules used was 1, and the adsorption energy of each IID-QAS molecule on the surface of copper was calculated, as shown in Table 3.

From the data in the table, the adsorption energy of the molecules on the copper surface in the following order: IID-C4-QL < IID-C4-DMAP < IID-C4-Bpy < IID-C4-py < IID-C4-IMD, the smaller the value of adsorption energy, the simpler the adsorption occurring, and the stronger the bonding of the molecules on the copper surface. The adsorption state of IID-QAS on the copper surface is nearly completely adhered to the copper surface in Fig. 10, which indicates that IID-QAS has a good adsorption pattern with a large adsorption area, among which IID-C4-QL has the lowest adsorption energy, and its bonding with copper atoms is more stable, so that its inhibitory properties can be better utilized. Combined with quantum chemical calculations, it can be concluded that IID-C4-QL has the best adsorption and inhibition properties in terms of molecular structure and can be used as a high-efficiency leveler, so it will be put into the next step of the actual copper plating test to explore the optimal amount of its addition.

3.5. Electroplating measurements

Through-hole copper plating experiments were carried out by adding IID-C4-QL as a leveler to the plating solution, the concentration of components of plating solution formulations (a–e) is shown in Table 4. Fig. 11 shows the cross-section of copper through holes obtained with different plating solutions using a metallurgical microscope, and the TP value of bath a is only 65.8% with poor filling efficiency, which will lead to nonuniform thickness of the copper layer in the orifice and hole due to the uneven distribution of the current density. Bath b had added SPS and PEG, and the TP value was increased to 86.27%. In baths c, d, and e, IID-C4-QL was added, and it could be observed that the thickness of the through-hole plating layer was gradually uniform with the concentration increasing, and the TP value of bath d reached up to 120.77%, which indicated that 1.5 ppm of IID-C4-QL had excellent filling performance.

In bath e, the concentration of IID-C4-QL was increased to 2 ppm, but the filling efficiency did not increase which the TP

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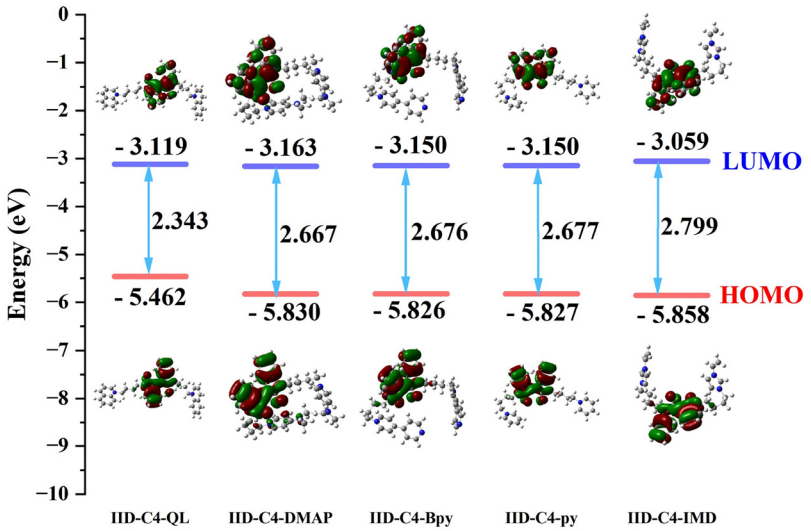


Fig. 8 Distributions and orbital energy values of HOMO orbital and LUMO orbital of IID-QAS.

Table 2 Orbital distribution and band gap of HOMO and LUMO in IID-QAS

Leveler	E_{HOMO} (eV)	E_{LUMO} (eV)	ΔE (eV)
IID-C4-QL	−5.462	−3.119	2.343
IID-C4-DMAP	−5.830	−3.163	2.667
IID-C4-Bpy	−5.826	−3.150	2.676
IID-C4-py	−5.827	−3.150	2.677
IID-C4-IMD	−5.858	−3.059	2.799

value decreased to 116.74%. There are only two reasons for the decrease of the TP value; one is because of the increase of the added amount, although there is convection-dependent adsorption behavior, the adsorption site of the leveler on the surface of the orifice is close to saturation, and it turns to adsorb inside the pore, which leads to the decrease of the thickness of the copper layer inside the pore; and the other is because of the large amount of the addition of IID-C4-QL, which has a competitive adsorption effect with the accelerator SPS. The adsorption sites of IID-C4-QL, which are mainly adsorbed on pore surface, are occupied by SPS, which weaken the inhibition performance on the pore surface, and increase the thickness of the copper layer, resulting in the increase of the TP value. The thickness of the copper layer inside the holes

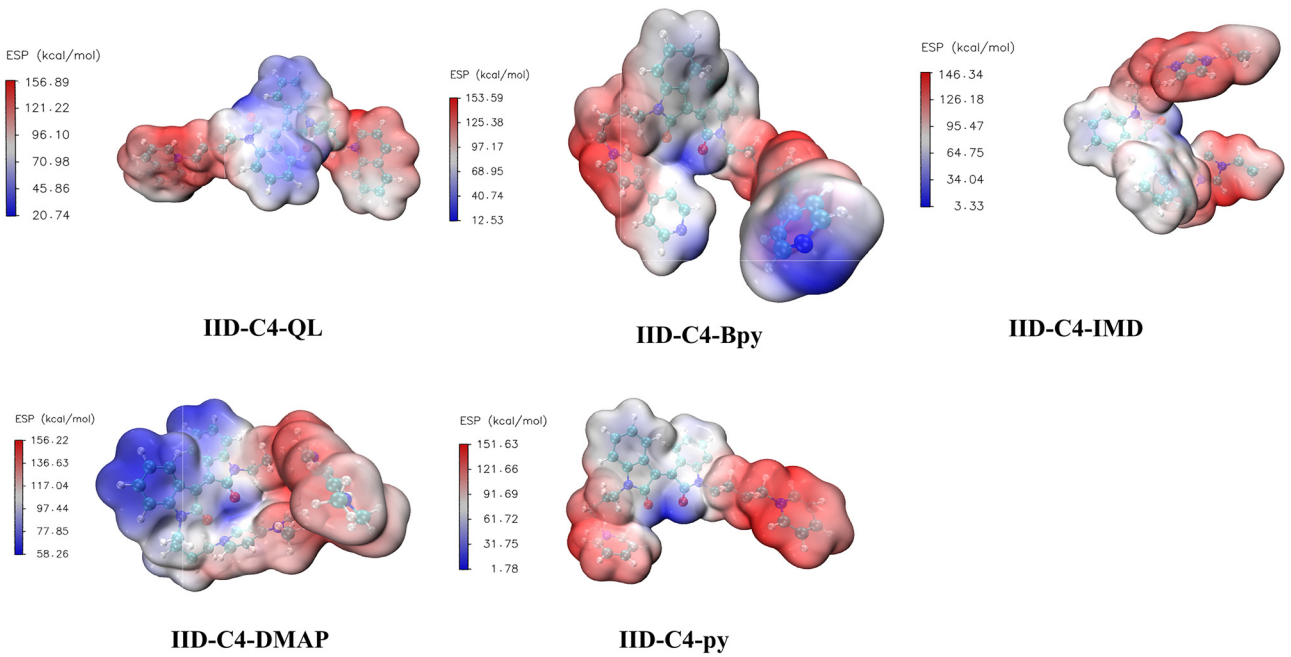


Fig. 9 The electrostatic potential (ESP) distribution on the surface of IID-C4-QL molecules.

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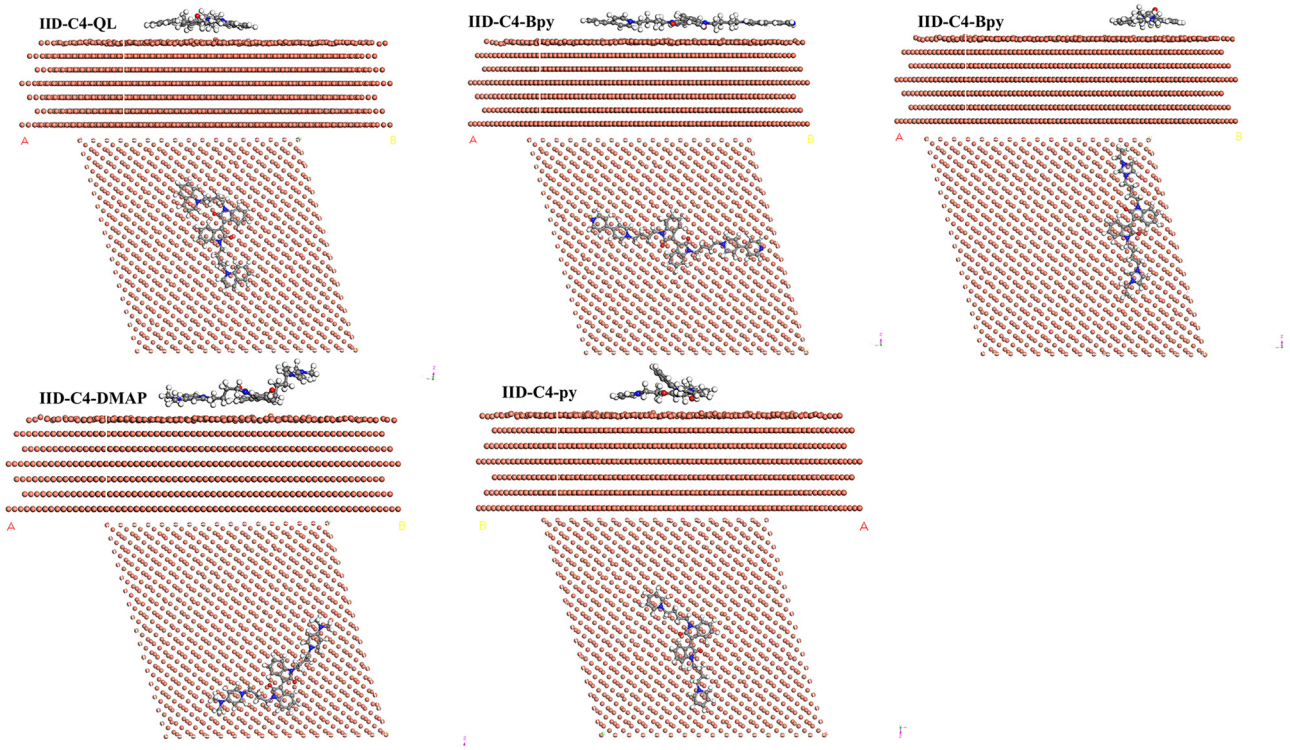


Fig. 10 Molecular dynamics simulation adsorption of IID-QAS on copper surface.

Table 3 Adsorption energy of IID-QAS molecular simulation adsorption

Leveler	E_{total} (kcal mol^{-1})	E_{Cu} (kcal mol^{-1})	$E_{molecule}$ (kcal mol^{-1})	E_{ads} (kcal mol^{-1})
IID-C4-QL	−61 403.759	−61 569.902	563.584	−397.442
IID-C4-DMAP	−61 678.084	−61 672.593	333.790	−330.281
IID-C4-Bpy	−61 577.689	−61 609.365	338.640	−306.964
IID-C4-py	−65 230.053	−65 171.103	226.273	−285.225
IID-C4-IMD	−61 779.175	−61 698.320	143.183	−224.038

Table 4 TP value measurement data

	Cl^- (ppm)	SPS (ppm)	PEG (ppm)	IID-QAS (ppm)	TP value (%)
Bath a	50	0	0	0	62.8
Bath b	50	1	200	0	86.27
Bath c	50	1	200	1	117.55
Bath d	50	1	200	1.5	120.77
Bath e	50	1	200	2	116.74

in Fig. 11 did not change much, but the thickness of the copper layer on the surface of the holes increased significantly compared with that of the bath d, so the main reason is the second one.

SEM was chosen to observe the surface morphology of the copper plating layer and the results are shown in Fig. 12. The surfaces of the copper layers corresponding to baths a and b with only PEG and SPS added were relatively rough; however, the copper layer was gradually flattened and the sizes of the copper crystal particles were also gradually refined in baths c and d. The bath d with the highest TP value, obtained the most flattened surface of the copper layer.

Although the surface of the copper layer corresponding to bath e is still flat, a lot of voids can be seen in the enlarged Fig. 12(f). Combined with the situation that the TP value decreased with the addition of 2 ppm of IID-C4-QL, the place where the voids appear is the distribution site where the IID-C4-QL is concentrated. The area around the hole is the area where SPS and IID-C4-QL compete for adsorption, and the excess leveler collects in the hole due to the occupation of adsorption sites by SPS, which leads to the high inhibition performance of the hole and the creation of the sunken hole. Therefore, the SEM results also show that 1.5 ppm is the optimal additive level, and that adding more than 1.5 ppm will result in the "cavity phenomenon".

According to the above experiments, the leveling mechanism of IID-C4-QL can be obtained as shown in Fig. 13, electrochemical analysis and theoretical calculations indicate that PEG and leveler adsorbed on the surface of the through-hole to inhibit copper deposition, while SPS and leveler compete for adsorption on the surface of the through-hole, and the competition for adsorption is intensified when the leveler is in excess, and the adsorption sites of the leveler are reduced, and piled up, so that the copper layer is thickened and the "cavity phenomenon" appears. In the copper electroplating measurements show that copper layer is thickened and the "hollow phenomenon" occurs.

XRD testing is shown in Fig. 14, the dominant crystal orientation of bath a without additives is (220), there is a small

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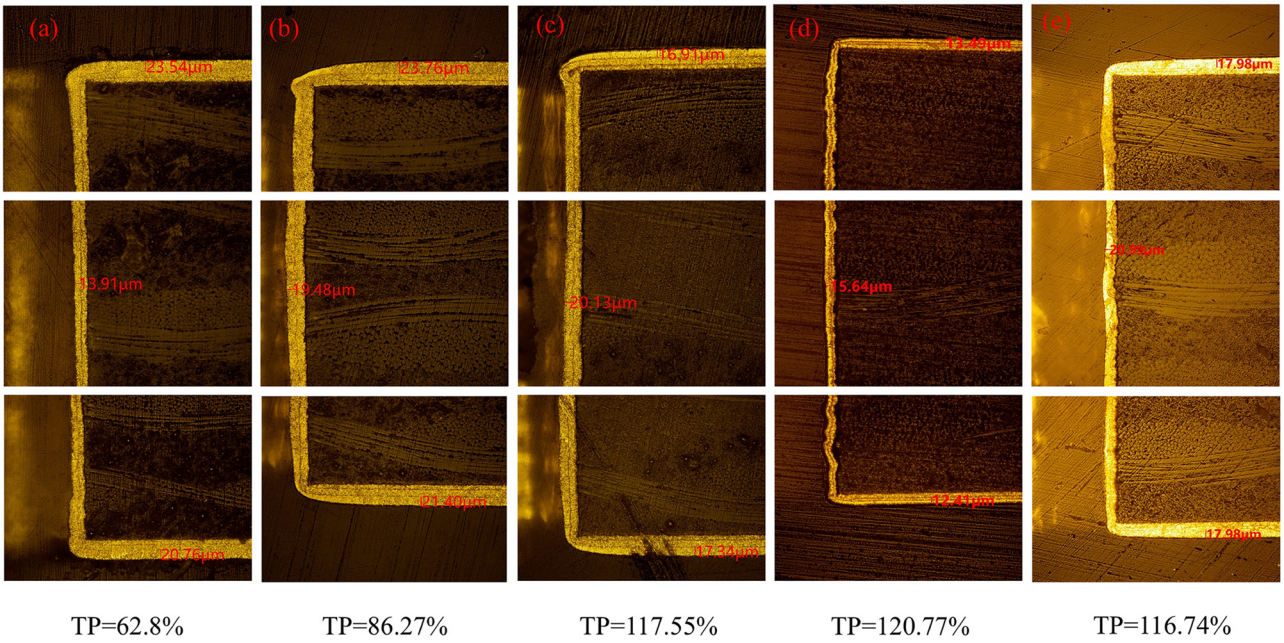


Fig. 11 Cross-section of plating through-hole corresponding to the plating formula (a)–(e).

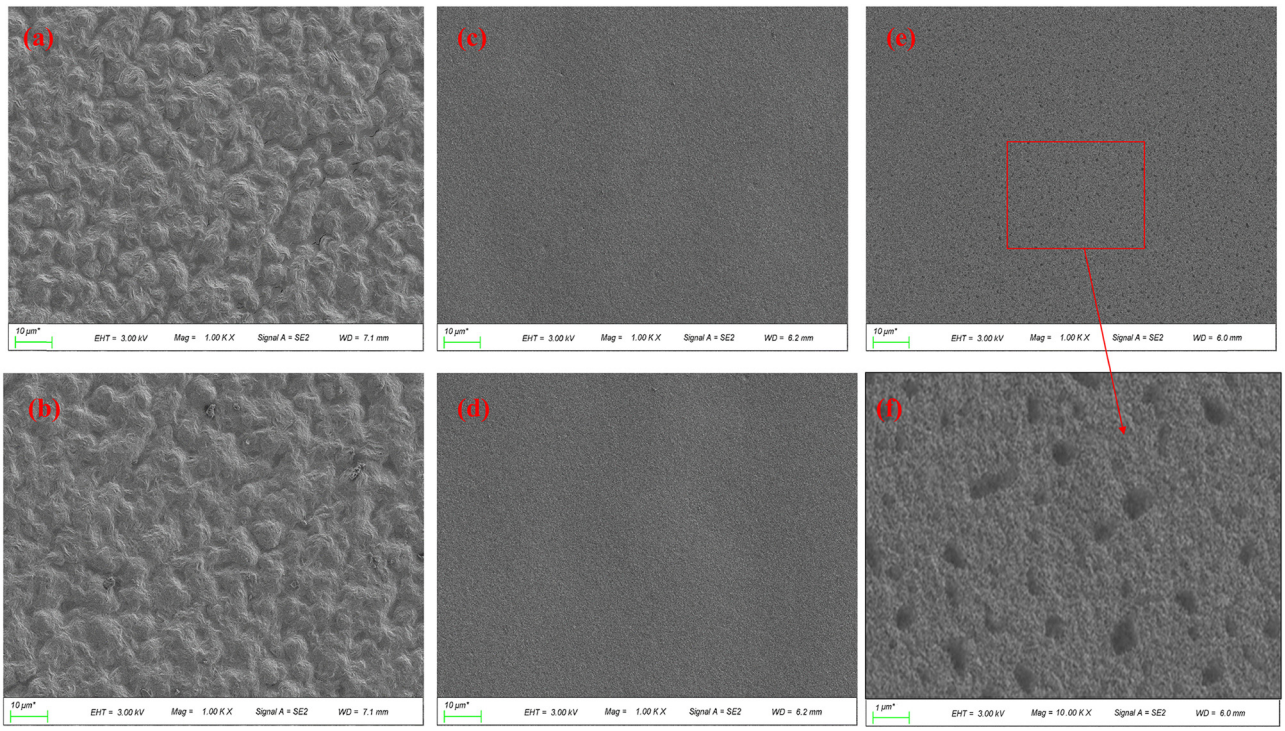


Fig. 12 Scanning electron microscope (SEM) image of the surface of the copper plate (magnification: 1000); figures (a)–(e) represent the surface morphology of copper plates obtained by electroplating in baths (a–e), respectively. (f) Partial enlarged view of (e) at 10 000 magnification.

intensity of (111) and (200) orientation, after the addition of IID-C4-QL crystal index (111) instead of (220) as the dominant crystal orientation. Therefore, the leveler greatly improves the transform efficiency of the conversion of the crystalline plane orientation to (111), there is no share of the intermediate crystalline plane orientation (200), and the intensity of the peaks of the dominant crystalline plane orientation is enhanced with the increase of the addition of the leveler.

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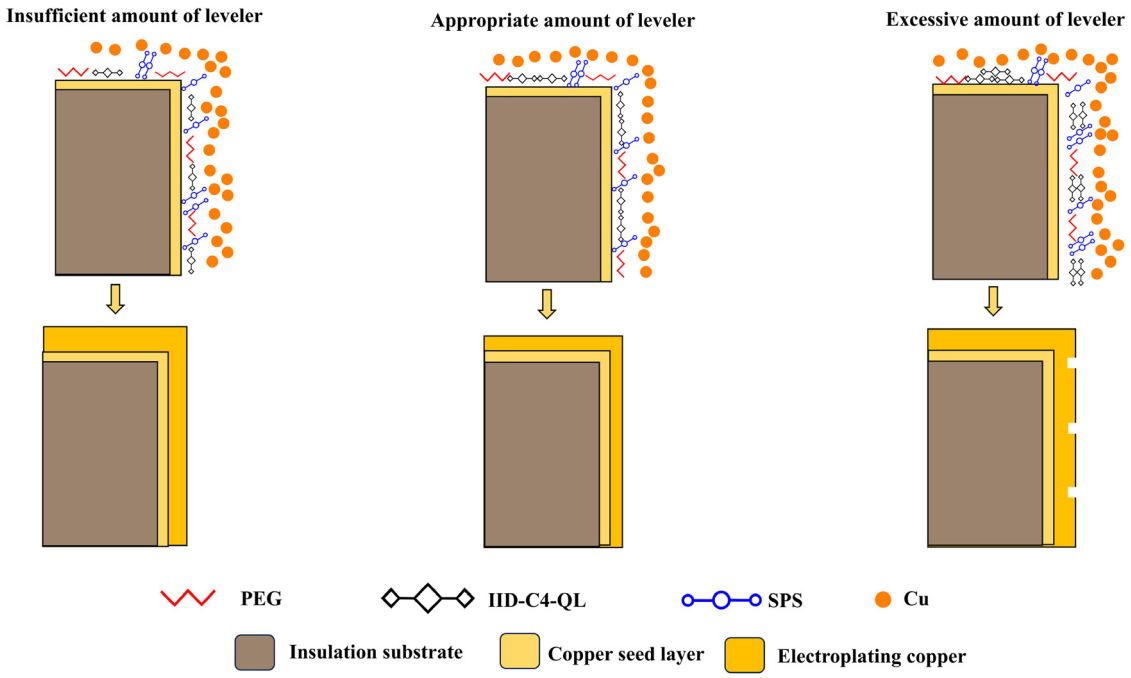


Fig. 13 Mechanism of three additives in the copper plating process.

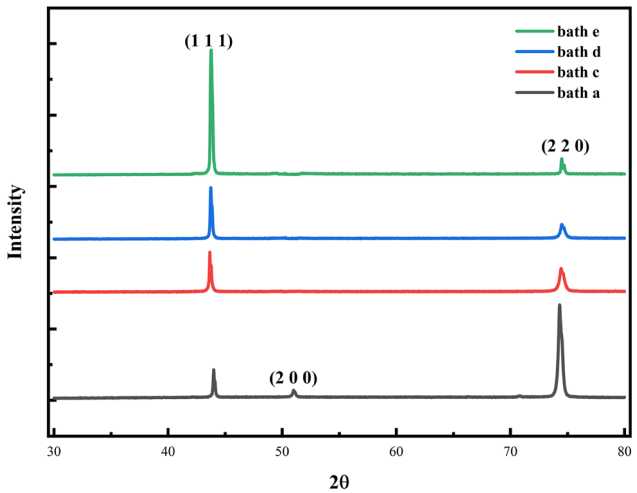


Fig. 14 XRD patterns of copper-plated films obtained by electrodeposition.

4. Conclusions

In summary, five isoindigo quaternary ammonium salt derivatives were synthesized in this paper using isoindigo as the parent body to explore their application as levelers in PCB through-hole plating. The results of electrochemical experiments show that five IID-QAS exhibit different degrees of copper deposition inhibition, among which IID-C4-QL has the best inhibition performance and strong convective adsorption ability. Theoretical calculations and molecular simulations show that IID-C4-QL has the strongest adsorption capacity among the five compounds. In the actual copper plating test, the highest TP value of 120.77% can be obtained with the addition of IID-C4-QL at 1.5 ppm, and the SEM results show that IID-C4-QL can obtain the flattest layer at the same concentration, while the addition of more than 1.5 ppm will compete with SPS for adsorption, resulting in the "cavity phenomenon" and decrease of the TP value of the layer. XRD results show that the addition of IID-C4-QL transforms the crystallographic orientation of the copper layer from (220) to the most stable (111) crystallographic plane. The present work enriches the study of isoindigo on this field and exploits the great potential and scope of isoindigo derivatives for acid plating applications.

Author contributions

Ruiqi Shan: conceptualization, investigation, resources, data curation, writing – original draft, writing – review & editing. Chenhao Qian: conceptualization, investigation, resources, data curation, writing – original draft. Xuyang Li: conceptualization, data curation, writing – original draft. Xin Chen: data curation, writing – review & editing. Limin Wang: supervision, project administration.

Conflicts of interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data are available upon request from the author or SI. The information about the synthesized compounds is available in

the SI linked to this article. Supplementary information is available. See DOI: https://doi.org/10.1039/d5tc01743b.

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